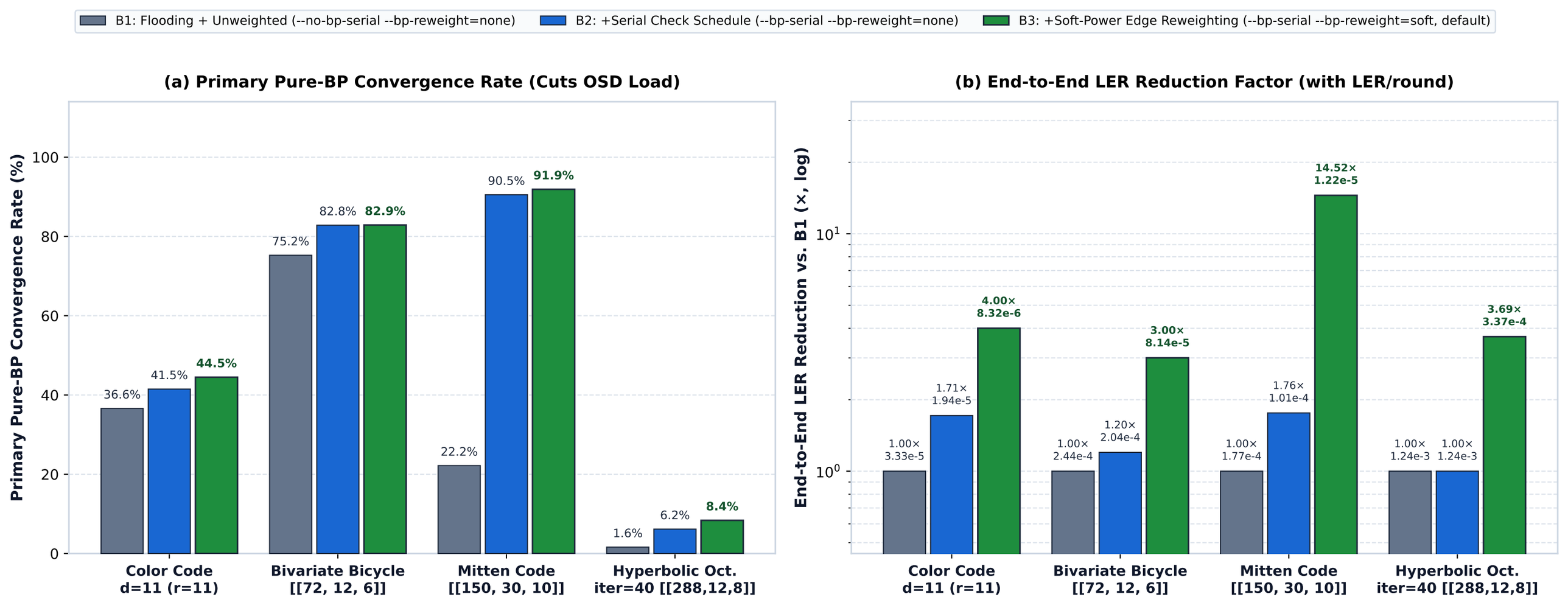
B1: Flooding + Unweighted (--no-bp-serial --bp-reweight=none)
B2: +Serial Check Schedule (--bp-serial --bp-reweight=none)
B3: +Soft-Power Edge Reweighting (--bp-serial --bp-reweight=soft, default)
(a) Primary Pure-BP Convergence Rate (Cuts OSD Load)
Primary Pure-BP Convergence Rate (%)
0
20
40
60
80
100
36.6%
41.5%
44.5%
75.2%
82.8%
82.9%
22.2%
90.5%
91.9%
1.6%
6.2%
8.4%
Color Code d=11 (r=11)
Bivariate Bicycle [[72, 12, 6]]
Mitten Code [[150, 30, 10]]
Hyperbolic Oct. iter=40 [[288,12,8]]
(b) End-to-End LER Reduction Factor (with LER/round)
End-to-End LER Reduction vs. B1 (×, log)
10^0
10^1
1.00× 3.33e-5
1.71× 1.94e-5
4.00× 8.32e-6
1.00× 2.44e-4
1.20× 2.04e-4
3.00× 8.14e-5
1.00× 1.77e-4
1.76× 1.01e-4
14.52× 1.22e-5
1.00× 1.24e-3
1.00× 1.24e-3
3.69× 3.37e-4
Color Code d=11 (r=11)
Bivariate Bicycle [[72, 12, 6]]
Mitten Code [[150, 30, 10]]
Hyperbolic Oct. iter=40 [[288,12,8]]